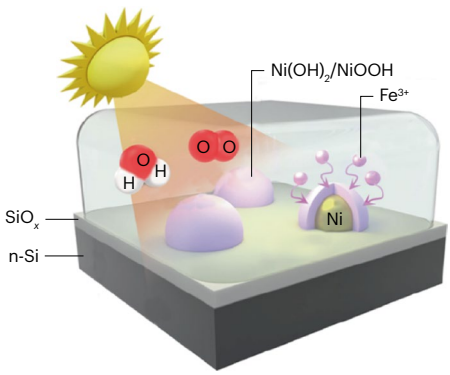
Photoelectrocatalysis

https://doi.org/10.1038/s41929-026-01503-0

A holistic view of PEC



In the field of electrocatalysis it is well-established that Fe-incorporation into Ni-based catalysts — whether intentional or due to impurity — has a profound impact on their activity in the oxygen evolution reaction (OER). Similar observations have been made for photoelectrochemical (PEC) analogues with Ni oxyhydroxides deposited as cocatalysts on semiconductor photoanodes. However, the enhancing effect of Fe impurities on such PEC systems has often been assumed to be electrocatalytic in origin while the junction energetics of these metal–insulator–semiconductor composites has been largely ignored. Now, Pattanasattayavong, Loget and co-workers demonstrate that synergistic electrocatalytic and photovoltaic effects caused by Fe doping on Si photoanodes modified with Ni nanoparticles (NPs) are essential to achieve the previously reported enhanced photovoltages of up 500 mV.

A suite of experiments including photoelectrochemical studies, elemental mapping and work function measurements demonstrated that Fe incorporation from the electrolyte into the shell of Ni NPs alters the surface energetics of the n-Si/SiO$_x$/NiNP composite photoanodes (pictured). A NP size dependence study revealed that sufficiently small NPs are required for these changes to result in an enhanced photovoltage. Furthermore, there is a requirement for Fe to be present in the electrolyte while PEC OER is occurring to prevent leaching and maintain the enhanced performance. Overall, these results point to some specific experimental conditions and catalyst design criteria (optimized NP size and electrolyte Fe concentration) in order to take best advantage of the Si/NiO$_x$ type photoanode.

Preliminary experiments with Co indicate that such effects may be generalizable to other metals and perhaps other semiconductor–metal interfaces. More broadly, this study points to the need for a more holistic approach to understanding semiconductor-cocatalyst photoelectrodes that incorporates both electrocatalytic and photoelectrochemical considerations.

Benjamin Martindale
Nature Catalysis

Original reference: *J. Am. Chem. Soc.* **148**, 5508–5519 (2026)

CREDIT: CC BY 4.0